

the fine-tuning argument

1. $Pr(e \mid \text{design}) = 0.5$
2. $Pr(e \mid \text{non-design}) = \frac{1}{10^{120}}$
3. $P(\text{design}) = P(\text{non-design}) = 0.5$.
4. Bayes' theorem.
5. $P(\text{design}) \approx 1$ (1,2,3)
6. If the universe was created by an intelligent designer, then God exists.

C. $P(\text{God exists}) \approx 1$.

Let's look at one last objection:

The multiverse objection

There are very many — perhaps infinitely many — distinct universes, which can have different fundamental physical constants.

Surely, this objection goes, if there were enough universes, then no matter how improbable it is that one of them would have constants that fall in the life-permitting range, it is not so improbable that some would. This would then be a way of rejecting premise (2) of the argument.

Our situation would then be somewhat analogous to the position of someone who wins the lottery. The odds of **that** person winning the lottery were very small; but if enough people buy tickets, it is not so improbable that someone wins.

It would be unreasonable for the winner to infer that the lottery was rigged in her favor; just so, it would be unreasonable for us to assume that our universe was designed.



fine-
tuning of the
universe

A solid red circle with a subtle drop shadow, centered on a white background. Inside the circle, the text "Bayes' theorem" is written in a white, sans-serif font, centered horizontally and vertically.

Bayes'
theorem



the
argument



objections

Let's consider a new way of questioning the probability assignment:

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Enough about probability (for now). Let's turn now to some results from

follows.

contemporary physics which will be important to the argument which

A first question: what does the theory provide by physics include?

“The standard model of physics presents a theory of the electromagnetic, weak, and strong forces, and a classification of all known elementary particles. The standard model specifies numerous physical laws, but that's not all it does. According to the standard model there are roughly two dozen dimensionless constants that characterize fundamental physical quantities.” (Hawthorne & Isaacs, “Fine-tuning fine-tuning”)

These 'dimensions' will be our focus.

One such constant is the cosmological constant, which measures the

energy density of empty space.

The cosmological constant is just a number which (as far as we know)

could have different values consistent with the laws of physics.

One thing that the standard model of physics gives us is a measure of how

(like the cosmological constant) would fall in a certain range.

likely it is, given the laws of the nature, that the fundamental constants

life to have evolved. For example, if there were nothing but hydrogen, it is

hard to see how life could have evolved. If there were no planets, it is hard

to see how life could have evolved.

We can make certain plausible assumptions about what it would take for

constant has a value which would permit the evolution of life.

Given assumptions such as these, we can look at what the standard model

of physics tells us about how likely it is that, for example, the cosmological

“Physicists have determined the (approximate) values of the fundamental constants by measurement.

(There's no way to derive the values of the fundamental constants from other aspects of the standard model. Any quantities that could be so derived wouldn't be fundamental.) Still, the underlying theory favored some sorts of parameter-values over others. ... Physicists made the startling discovery that -- given antecedently plausibly assumptions about the nature of the physical world -- the probability that a universe with general laws like ours would be habitable was staggeringly low.”

supporting value, given the laws of nature, is very low.

This is the claim that the cosmological constant having a life-

We should emphasize just how small we take the life-permitting parameter values to be according to the physically-respectable measures. “Small” here doesn’t mean “1 in 10,000” or “1 in 1,000,000”. It means the kind of fraction that one would resort to exponents to describe, as in “1 in 10 to the 120”. The kind of package that we have in mind tells us that only a fantastically small range is life permitting.

nature, is

cosmological constant falling in a life-supporting range, giving the laws of

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1



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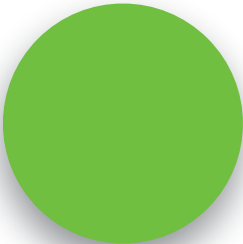


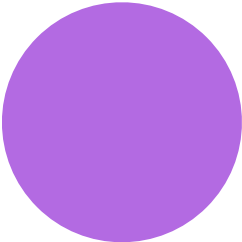
fine-
tuning of the
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Bayes'
theorem











nature, is

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probability theory into an argument for the existence of God.

Let's now ask how to turn these facts about current physics and

in the turn, so we can consider two different hypotheses about the origins

of the universe.

Much as we considered two different hypotheses about the number of balls

Let the design hypothesis be the hypothesis that the universe

created by an intelligent designer. Let the non-design hypothesis be the

hypothesis that it was not.

Let e be the fact that the cosmological constant has a

value in the life-permitting range.

Then what do we know about relevant probabilities?



$$Pr(e \mid \text{non-design}) = \frac{1}{10^{120}}$$



$$Pr(e | \text{design}) \equiv 0.5$$

Bayes' theorem

$$P(h|e) = \frac{P(h)*P(e|h)}{P(e)}$$

numbers.

With this information in hand, figuring out the probability of the non-

design hypothesis given evidence is that just plugging in the



$$Pr(\text{non-design}) = 0.5$$



$$Pr(\text{design}) = 0.5$$



$$Pr(e) = 0.25$$



$$Pr(\text{non-design} \mid e) = \frac{0.5 * \frac{1}{10^{120}}}{0.25} = \frac{2}{10^{120}}$$

PowerBar!

It is difficult to think about numbers as large as the denominator of this fraction. But

to give you some idea: the odds of winning Powerball are about 1 in 300 million. Now



consider the odds of winning Powerball one trillion times in a row. Call that a “super

Now consider the odds of winning a super Powerball one trillion times in a row. Call

that a 'super duper Powerball.'

The odds of this happening are about $1/10^{44}$ — so much, much higher than the

odd of the universe being life-permitting by chance.

Now consider the odds of winning a super Powerball one trillion times in a row.

the non-design hypothesis being true are vastly lower than the chances of winning a

probability of 0.5 to the non-design hypothesis, you should think that the chances of

This means that if you simply take the physics at face value, and begin by assigning a

superduperpowerball a trillion times in a row.

By contrast, the probability of the design hypothesis is very close to 1. It is, in fact, 1.

minus the very small number we're just discussing:



$$Pr(\text{design} \mid e) = 1 - \frac{2}{10^{120}}$$

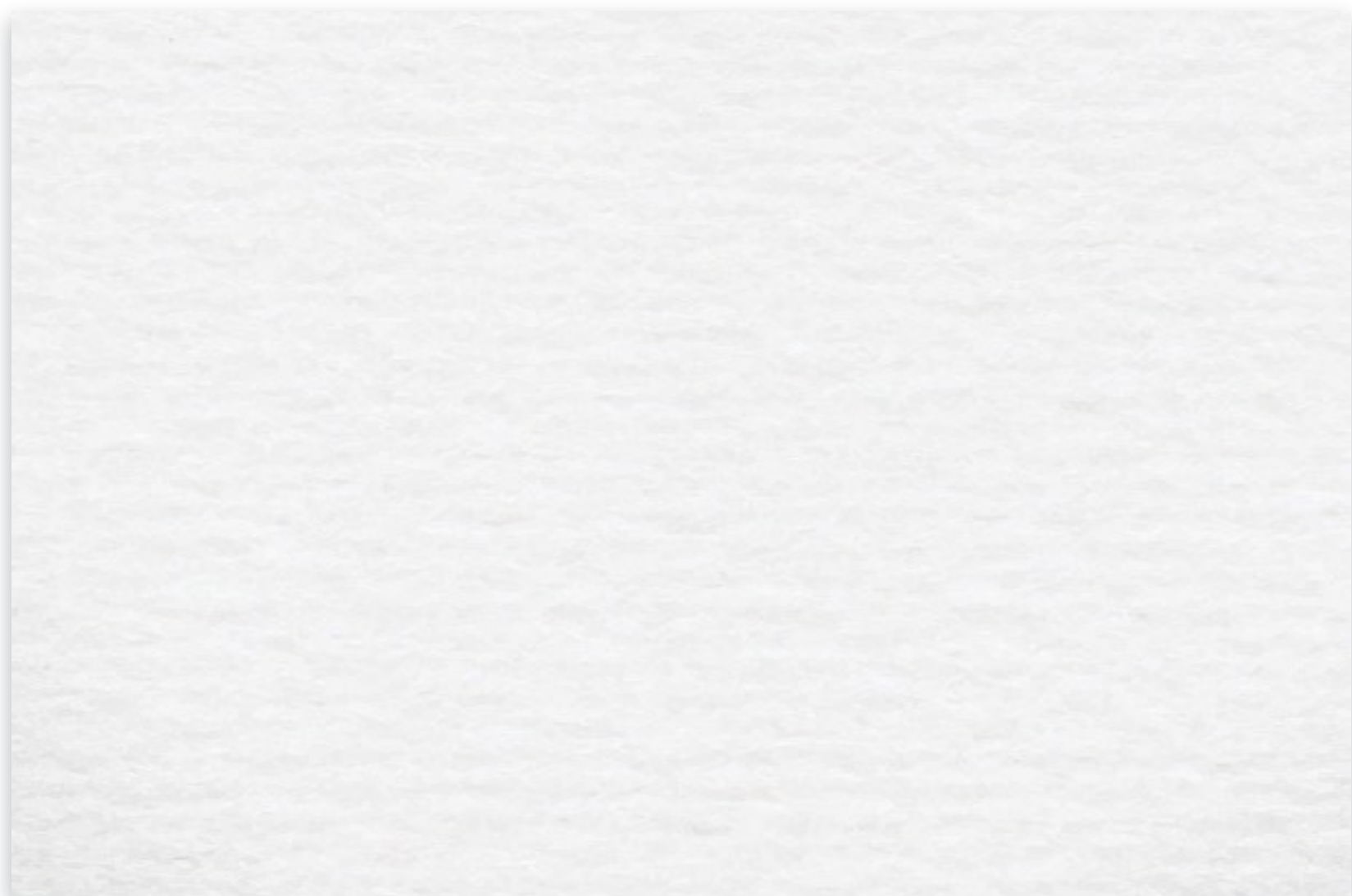
This about as close to certainty as it is possible to get.



the
argument



objections



2



P



$$3. P(\text{design}) = P(\text{non-design}) = 0.5.$$

1, 2, 3, 4

4. Bayes', theorem.



$$7. P(\text{design}) \approx 1. \quad (5, 6)$$

5



P

intelligent designer, then God exists.

6. *Life exists.*

$$\text{C.P.}(\text{God exists}) \approx 1. (7, 8)$$

8. If the universe was created by an

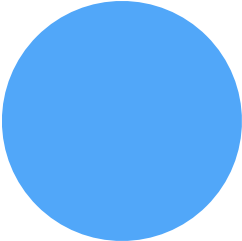
$$(life/death) \equiv 0.5$$

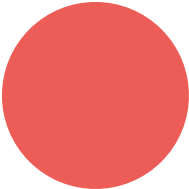
$$(1ife | non-design) = \frac{1}{10^{120}}$$

$$(\text{design} / \text{life}) \approx 1$$



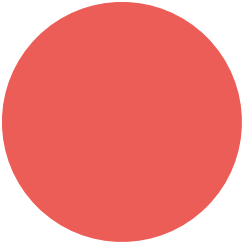








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The convinced atheist objection

The hypothesis that an intelligent designer exists is really, really unlikely in the first place, so premise (3) is false.

Skepticism about physics

The physics is wrong (or at least
might be).

The anthropic objection

We could never observe the falsity of the claim that the constants permits life since, if it were false, we would not exist to observe it.

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